Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/26/2008
PI Name:	Roux, Stanley Ph.D.		
Project Title:	Early development of fern gametophytes in microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	95-OLMSA-02
Start Date:	01/10/2001	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (12/06)		
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COI Name (Institution):			
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	The main goals of this project are to further document the effects of microgravity on gene expression and polarity development in single spore cells of the fern Ceratopteris richardii, and to test the functional significance of these effects. The spore cells show a characteristic gravity response on earth, which is that gravity directs their developmental polarity. The first microscopic evidence that gravity has fixed the polarity of the cell is that the centrally located nucleus migrates downward, setting up an asymmetric first cell division. Experiments on Shuttle mission STS-93 provided initial documentation of the effects of microgravity on polarity development, nuclear migration, and the pattern of gene expression in fern spores. Results showed that nuclear migration and polarity development both occurred but were randomly directed in microgravity, and that the pattern of gene expression was different in microgravity relative to ground controls for a small percentage of genes. The STS-93 experiments did not have an on-board 1-g control, and a		

Task Description:	low percentage of the spores were induced to germinate prematurely during handling before launch. Originally, the main goal of this project was to correct both of these deficiencies in a re-flight, and, additionally, to more comprehensively document the effects of microgravity on gene expression, using microarray technology. However following the severe reduction in fllight opportunities after the Columbia tragedy, NASA decided that a flight opportunity for this experiment was unlikely and the main focus of this project was switched to research on two topics tightly linked to the original goal: 1) Production and probing of microarrays of the 5000 cDNAs in the Ceratopteris EST library. 2) Further characterization of the rapid reversal of the trans-cell calcium current in single-celled Ceratopteris spores when these spores are turned upside down. We have made significant progress on both of these topics during the past year and these are detailed in the report below. In addition, a new microchip technology was developed to measure very rapid responses of single cells to microgravity achieved for brief periods during parabolic flight of NASA's DC9 aircraft, and exciting new data were generated with this new technology. Major findings from this project are summarized below under three headings. This study represents a rare, if not unique, instance in which the results will provide both an extensive analysis of gene expression changes induced by the microgravity environment in single cells and detailed information on the molecular components that are among the most rapidly altered by the gravity stimulus. This kind of gene-level and cell-level information is critical for formulating and clarifying ideas on the most basic biological mechanisms that are affected by the gravity stimulus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beside providing new insights into the cellular basis of gravity responses, the data obtained from this project will be more broadly valuable for studies of cell polarity development and of mechanisms of calcium-mediated signaling. This research does not directly seek to understand a disease or malady that affects humans on Earth and/or in space, nor does it seek to develop new therapeutics for alleviating symptoms of a malady on Earth. However, the findings will help to understand the growth of plants better, and since plants are a crucial source of food for humans, this research does seek to understand the malady of malnutrition. Also, this research does yield a new understanding of basic biological processes, specifically the processes of plant growth and the cellular mechanisms whereby gravity can affect the developmental polarity in cells. Further, this research points to a real role of gravity in regulating growth and development of plants on Earth and thus reveals potential problems in achieving normal growth and development of plants in space. Finally, the health of humans is inextricably linked to the ability to control and continuously improve the growth of plants. This, in turn, requires an improved understanding of the molecular mechanisms that control growth in plants. The accomplishments of this research contribute to that improved understanding, and thus indirectly benefit humans.
	The major advances we made on this project are summarized below under three headings. 1. Major Gene Expression Changes Occur While Gravity Is Directing the Polarity of Fern Spore Development DNA Microarray technology has provided a method of monitoring expression profiles of almost any biological system during developmental changes under a variety of different conditions. We chose to use the microarray approach to get a global assessment of gene expression changes that are induced by the microgravity environment of the Shuttle mid-deck during flight in outer space. We mastered several technical hurdles to produce uniform microarrays of 3,207 genes (cDNAs) taken from an Expresed Sequence Tag library that had been made from RNA isolated from single spore cells of Ceratopteris 20 h after their germination was induced by light. We tested the reproducibility of the system and to get a rigorous assessment of the gene expression changes that occur developmentally at 1-g during the first 48 h of the germination process, we carried out a study of the expression changes that occur during this period, which includes the time during which gravity determines the polarity of spore development. For this analysis, we used thoroughly replicated microarray data analyzed by a Bayesian model for determining statistically significant changes. We gathered data from 34 array hybridizations, which included at least 8 replications of 4 different developmental time point comparisons. Analysis of these data revealed over 900 genes with relative changes in transcript abundance over the first 48 hours of spore development. Within the first 24 hours about 1% of genes on the microarray were differentially expressed at least 2.5 fold. We also used real time quantitative RT-PCR to independently verify the expression changes observed in microarray analysis. After the analyses described above we obtained another 1200 ESTs to raise the total in our library to nearly 5000. Analysis revealed that in this library about 3930 unique genes were represented. In order to determine the EST library’s percent coverage of the actual population of genes expressed during spore germination, we estimated the number of unique genes present in the population by applying an approach used to estimate species richness in ecological studies. According to this estimate there are about 14,250 unique genes expressed 20 h after light initiation of spore germination, suggesting that the current 3930 unique genes represent approximately 27% of those sequences. The identity of the genes designated as unique was determined by BLAST analysis against the Arabidopsis proteome, yielding 2710 (ca. 70%) with significant homology (E-value = 1.0 × 10-10) to Arabidopsis proteins. Using the Gene Ontology terms of the matching Arabidopsis loci, functional and localization assignments were made for each of these Ceratopteris genes. The functional patterns seen in Ceratopteris spores were generally similar to those found in the Arabidopsis tissues sampled. A poster report on this work by the main hands-on investigator, Mari Salmi, won first prize for best graduate presentation, plant category, at the annual meeting of the American Society for Gravitational and Space Biology in November 2003. Later, this work was published in Plant Physiology (Salmi et al., 2005). Because the quantitative assays of transcript abundance measured in this work were internally consistent and reproducible, we could be confident to proceed to a global comparison of gene expression patterns at 1-g vs. micro-g, To do this we used RNA extracted from spores flown on STS-93 in a high-throughput evaluation of gene expression changes induced by space flight in a single cell, the germinating spore of Ceratopteris richardii, which exhibits both perception and response to gravity. cDNA microarray and qRT-PCR analysis of spores germinating in microgravity onboard NASA space shuttle flight STS-93 revealed changes in the mRNA expression of roughly 5% of genes analyzed. These gene expression changes showed some interesting similarities to those that occur during gravity perception and response in animal cells and multicellular plants. These findings contributed to a more thorough understanding of the impact of gravity on cellular growth and development, and provided insights into the adaptive strategies of individual cells in response to this force. A manuscript on this work has been submitted to Planta and is currently under review.

Task Progress:	2. Molecular Components of the Gravity-Directed Calcium Current in Ceratopteris: Characterization of a Membrane-Bound Calcium Pump During the period in which gravity determines their developmental polarity, germinating spores of the fern Ceratopteris richardii exhibit a bottom-to-top calcium current. This current is capable of rapid reorientation, reversing direction in less than 42 sec when the spores are turned upside down. The current is parallel to the subsequent axis of cell development and blocking the current with nifedipine disrupts the ability of gravity to direct the polarity of spore development. The entry of calcium into the bottom of the cell is presumably mediated by a channel-type transporter and its exit from the top of the cell by an ATP-dependent pump. The function of the channel(s) and pump(s) that drive the current must be rapidly modulated by gravity and understanding how gravity does this will require identifying these transporters and characterizing the mechanisms that control their function. Toward this end, we identified three cDNAs from a Ceratopteris EST library whose sequences predict that they encode Ca2+ATPase pumps. Two of them encode proteins that are most similar to calcium pumps known to be plasma membrane localized. The expression of one of these (CrACA1) appears relatively unchanged during the period when the gravity-directed current is strongest, i.e., the first 16 hours after germination is initiated by light, as judged both by microarray and quantitative RT-PCR assays. We raised antibodies to a 93-mer peptide corresponding to the large loop between transmembrane regions 4 and 5 of CrACA1, and demonstrated their specificity in immunoblot analyses. These investigations were important first steps in the identification of the pump(s) that drive the gravity-directed calcium current, and in the discovery of the mechanisms by which gravity controls pump function. A poster report on this work by the main hands-on investigator, Thomas Bushart, won first prize for best graduate presentation, plant category, and the annual meeting of the American Society for Gravitational and Space Biology in November 2004. It was published in Dr. Bushart's 2007 Ph.D. dissertation (University of Texas), and is included in a manuscript now being prepared for submission to a top journal. 3. Development of a A MEMS Fabricated Cell Electrophysiology Laboratory Biochip for In-silico Ca2+ Measurements, and Use of this Chip to Document that Changes in Gravity Rapidly Alter the Magnitude and Direction of a Ca2+ Current Across Fern Spores The first experiments to measure a gravity-directed calcium flux in Ceratopteris spore cells used a self-referencing ion-selective electrode system that sequentially measured the directional flux of calcium at various points around the spore (4, 5, 6). Although these experiments showed that the direction of the current inverted when the cells were rotated 180o, they did not reveal how quickly the current changed direction, so interpretation of whether gravity was directly or indirectly altering the activity of the transporters that drove the current was not possible. We upgraded the electrode system with computer controlled stepper motors to rotate individual spores and measured calcium fluxes at two of the three positions around the spore (bottom, top, and side) before and after 180 degree rotation of the spore, and confirmed that the gravity-directed reversed in less than a minute after the cell was inverted. Then, in collaboration with Marshall Porterfield (Purdue University), we also devised a microfluidic device that could measure the calcium current across 16 cells simultaneously and continuously in real time, and this permitted both an assay of gravity-directed changes in the calcium current in real time during spore rotation and an assessment of whether changes in the force of gravity altered the magnitude of the current. The design, fabrication, and testing of this microchip (referred to as a cell electrophysiology lab-on-a-chip (CEL-C)), and the interface instrumentation and software for measuring differential calcium concentrations were published in two separate reports (ul Haque et al., 2006; ul Haque et al., 2007). Using the microfluidic device, we demonstrated that the direction of this polar calcium flux is reversed in less than 30 sec when the cells are inverted, and the magnitude of the flux rises and falls rapidly in parallel with the rapidly changing g-forces during parabolic flight on the NASA DC9 aircraft. We found that the current was suppressed by either nifedipine (calcium channel blocker) or eosin yellow (calcium pump inhibitor), but only nifedipine disrupts the ability of gravity to direct cell polarity. Spores were verified to be expressing the gene and protein for a plasma membrane-type calcium pump. Our results indicated that gravity perception may be mediated by a mechanically activated ion channel in this plant cell, an idea consistent with previously proposed models of plant gravity perception. These findings have been summarized in a manuscript now being prepared for submission to a top journal.
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